THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

ISE 430/530: Manufacturing Systems and Facilities Design

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Steps in CRAFT

1. Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).

2. Find the cost of the initial layout by multiplying the from-to chart, unit cost matrix and the distance matrix.

3. Improve the layout by performing all-possible two or three-way exchanges (we just consider two-way exchanges here if not stated otherwise).

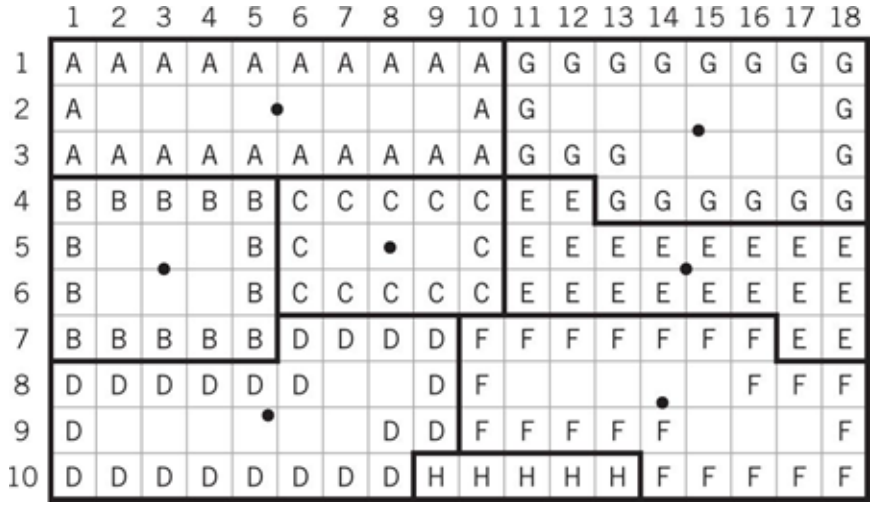
   - At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

4. If possible, make adjustments (“Massaging”) for departments layout to be practical.

CRAFT Example

- A manufacturing facility with 7 departments. Total available space is 72,000 ft2. We use 20'x20' grid as the unit area. For practical reasons, locations of receiving (A) and shipping (G) departments are fixed.

- From-to chart and current layout are given.



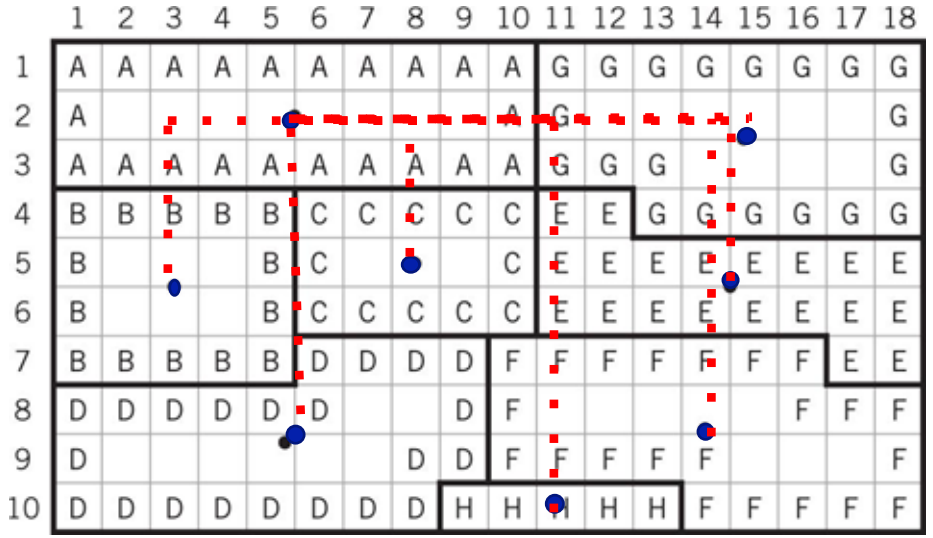
Space requirement: 70,000			FLOW							
Department Name	Area (ft^2)	# of Grids	A	B	C	D	E	F	G	H
1. A:Receiving	12,000	30	0	45	15	25	10	5	0	0
2 B:Milling	8,000	20	0	0	0	30	25	15	0	0
3 C:Press	6,000	15	0	0	0	0	5	10	0	0
4 D:Screw m/c.	12,000	30	0	20	0	0	35	0	0	0
5 E:Assembly	8,000	20	0	0	0	0	0	65	35	0
6 F:Plating	12,000	30	0	5	0	0	25	0	65	0
7 G:Shipping	12,000	30	0	0	0	0	0	0	0	0
8 H:Dummy	**2,000**	**5**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

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Steps in CRAFT

- Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).

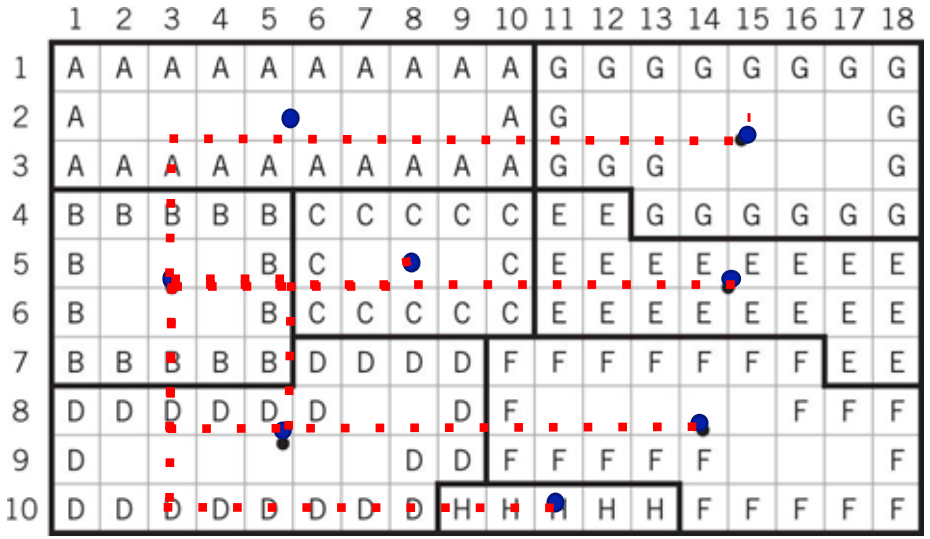
t	A	B	C	D	E	F	G	H
A		6	5.5	6.5	12.5	15	9.5	13.5
B								
C								
D								
E								
F								
G								
H								



CRAFT Example

- Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).

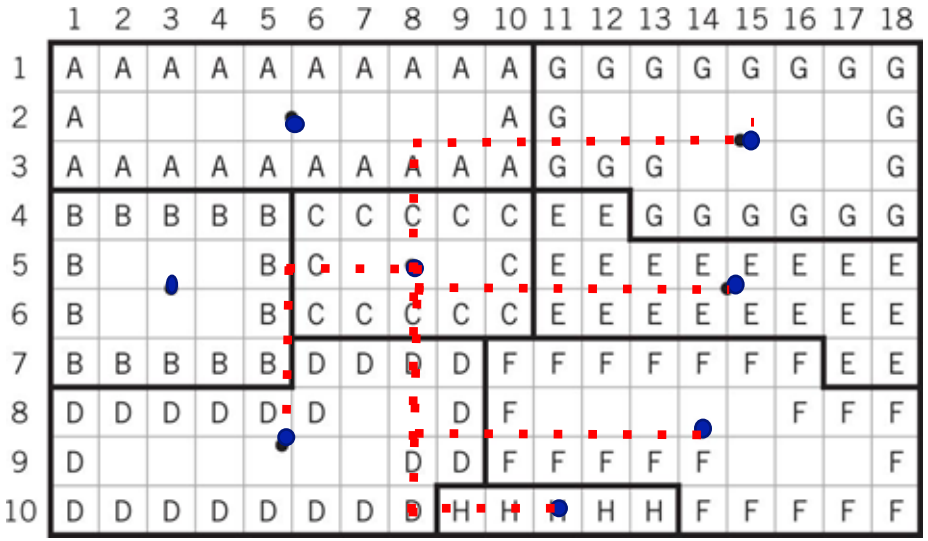
t	A	B	C	D	E	F	G	H
A		6	5.5	6.5	12.5	15	9.5	13.5
B			**5.5**	**5.5**	**11.5**	**14**	**15**	**12.5**
C								
D								
E								
F								
G								
H								



CRAFT Example

- Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).

t	A	B	C	D	E	F	G	H
A		6	5.5	6.5	12.5	15	9.5	13.5
B			5.5	5.5	11.5	14	15	12.5
C				6	7	9.5	9	8
D								
E								
F								
G								
H								



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CRAFT Example

- Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).

t	A	B	C	D	E	F	G	H
A	0	6	5.5	6.5	12.5	15	9.5	13.5
B	6	0	5.5	5.5	11.5	14	15	12.5
C	5.5	5.5	0	6	7	9.5	9	8
D	6.5	5.5	6	0	12	8.5	15	7
E	12.5	11.5	7	12	0	3.5	3	8
F	15	14	9.5	8.5	3.5	0	6.5	4.5
G	9.5	15	9	15	3	6.5	0	11
H	13.5	12.5	8	7	8	4.5	11	0

			FLOW							
Department Name	Area (ft^2)	# of Grids	A	B	C	D	E	F	G	H
1. A:Receiving	12,000	30	0	45	15	25	10	5	0	0
2 B:Milling	8,000	20	0	0	0	30	25	15	0	0
3 C:Press	6,000	15	0	0	0	0	5	10	0	0
4 D:Screw m/c.	12,000	30	0	20	0	0	35	0	0	0
5 E:Assembly	8,000	20	0	0	0	0	0	65	35	0
6 F:Plating	12,000	30	0	5	0	0	25	0	65	0
7 G:Shipping	12,000	30	0	0	0	0	0	0	0	0
8 H:Dummy	2,000	5	0	0	0	0	0	0	0	0

Total cost:

Steps in CRAFT

- Improve the layout by performing all-possible two or three-way exchanges. At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

If we were to exchange A-B, estimate new cost

t	A	B	C	D	E	F	G	H
A	0	6	5.5	6.5	12.5	15	9.5	13.5
B	6	0	5.5	5.5	11.5	14	15	12.5
C	5.5	5.5	0	6	7	9.5	9	8
D	6.5	5.5	6	0	12	8.5	15	7
E	12.5	11.5	7	12	0	3.5	3	8
F	15	14	9.5	8.5	3.5	0	6.5	4.5
G	9.5	15	9	15	3	6.5	0	11
H	13.5	12.5	8	7	8	4.5	11	0

			FLOW							
Department Name	Area (ft^2)	# of Grids	A	B	C	D	E	F	G	H
1. A:Receiving	12,000	30	0	45	15	25	10	5	0	0
2 B:Milling	8,000	20	0	0	0	30	25	15	0	0
3 C:Press	6,000	15	0	0	0	0	5	10	0	0
4 D:Screw m/c.	12,000	30	0	20	0	0	35	0	0	0
5 E:Assembly	8,000	20	0	0	0	0	0	65	35	0
6 F:Plating	12,000	30	0	5	0	0	25	0	65	0
7 G:Shipping	12,000	30	0	0	0	0	0	0	0	0
8 H:Dummy	2,000	5	0	0	0	0	0	0	0	0

CRAFT Example

- Improve the layout by performing all-possible two or three-way exchanges. At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

If we were to exchange A-B, estimate new cost

Perform all possible swapping between two or three “adjacent” departments

Total cost:
sum (flow * dist) = 2735 units

			FLOW							
Department Name	Area (ft^2)	# of Grids	A	B	C	D	E	F	G	H
1. A:Receiving	12,000	30	0	45	15	25	10	5	0	0
2 B:Milling	8,000	20	0	0	0	30	25	15	0	0
3 C:Press	6,000	15	0	0	0	0	5	10	0	0
4 D:Screw m/c.	12,000	30	0	20	0	0	35	0	0	0
5 E:Assembly	8,000	20	0	0	0	0	0	65	35	0
6 F:Plating	12,000	30	0	5	0	0	25	0	65	0
7 G:Shipping	12,000	30	0	0	0	0	0	0	0	0
8 H:Dummy	2,000	5	0	0	0	0	0	0	0	0

CRAFT Example

- See the excel for comparisons

"Code Craft: The Practice of Writing Excellent Code" – see the coding here.

t	A	B	C	D	E	F	G	H
A	0	6	5.5	6.5	12.5	15	9.5	13.5
B	6	0	5.5	5.5	11.5	14	15	12.5
C	5.5	5.5	0	6	7	9.5	9	8
D	6.5	5.5	6	0	12	8.5	15	7
E	12.5	11.5	7	12	0	3.5	3	8
F	15	14	9.5	8.5	3.5	0	6.5	4.5
G	9.5	15	9	15	3	6.5	0	11
H	13.5	12.5	8	7	8	4.5	11	0

Minimum cost occurs when E-F are swapped:
sum (flow X dist) = 2410 units

Notice:

1. These distances do not stay the same. Department areas are different, therefore centroids locations will change, causing distances to change.
2. The objective function reduction assessments are also estimates only.

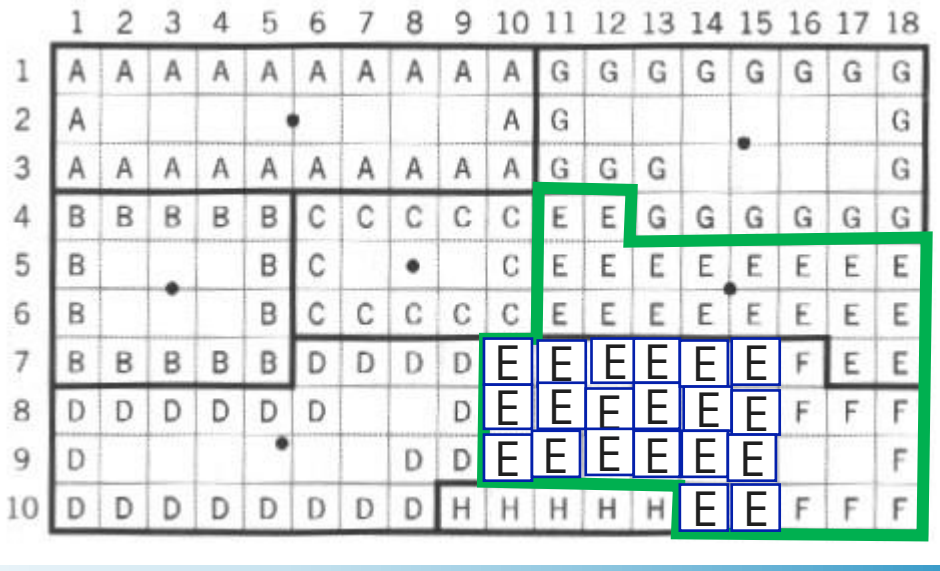
CRAFT Example

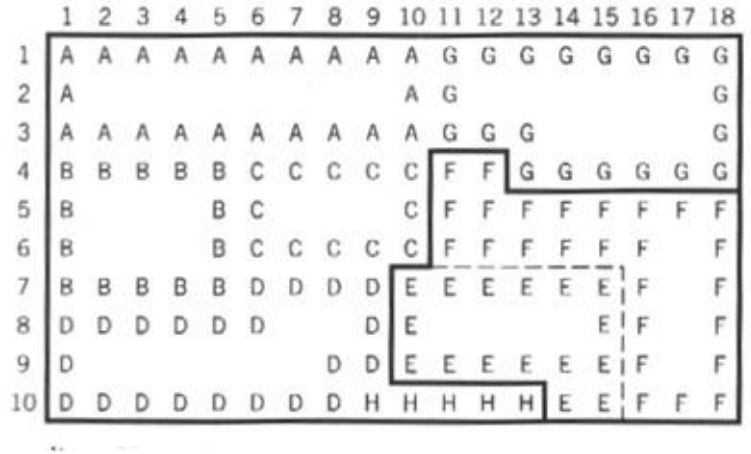
- Improve the layout by performing all-possible two or three-way exchanges. At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

Exchanging E(20)-F(30)

E-F are adjacent: we can draw a box containing only E and F

How to exchange? Strategy 1: Fill E grids (20) from leftmost column of what is currently F

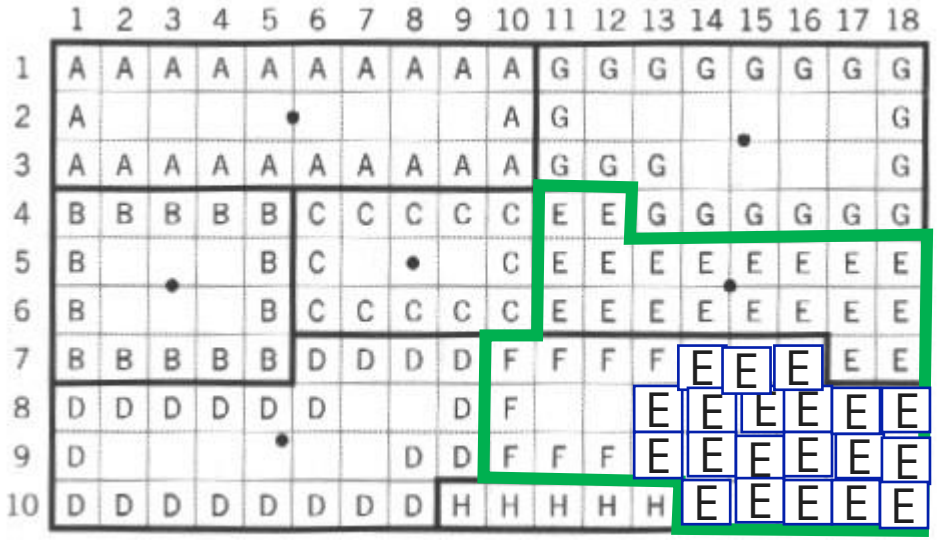




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CRAFT Example

- Strategy 2: Fill E grids (20) from rightmost column of what is currently F



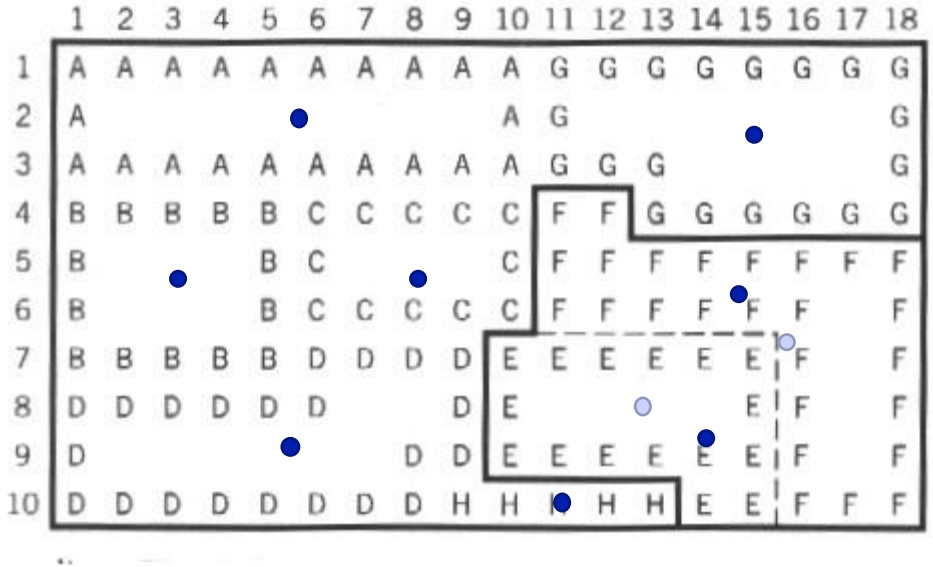
Notice that department shapes are poor. In fact, CRAFT often deteriorate as progress through iterations even if initial shapes are rectangular.

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CRAFT Example

- Improve the layout by performing all-possible two or three-way exchanges. At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

  Identify new centroids and update distance matrix and update the objective function



Estimate: 2735
Real cost: 3015

END of first iteration
Continue step 3

CRAFT Example

- Improve the layout by performing all-possible two or three-way exchanges. At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

Identify new departmental exchange that would result in minimum cost (estimate)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	A	A	A	A	A	A	A	A	A	A	G	G	G	G	G	G	G	G
2	A									A	G							G
3	A	A	A	A	A	A	A	A	A	A	G	G	G					G
4	B	B	B	B	B	C	C	C	C	C	F	F	G	G	G	G	G	G
5	B				B	C				C	F	F	F	F	F	F	F	F
6	B				B	C	C	C	C	C	F	F	F	F	F	F		F
7	B	B	B	B	B	D	D	D	D	E	E	E	E	E	E	F		F
8	D	D	D	D	D	D			D	E					E	F		F
9	D							D	D	E	E	E	E	E	E	F		F
10	D	D	D	D	D	D	D	D	H	H	H	H	H	E	E	F	F	F

END of second iteration
Continue step 3
No improvement can be made

B-C

Perform this for all possible swapping between two or three "adjacent" departments

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	A	A	A	A	A	A	A	A	A	A	G	G	G	G	G	G	G	G
2	A									A	G							G
3	A	A	A	A	A	A	A	A	A	A	G	G	G					G
4	C	C	C	B	B	B	B	B	B	B	F	F	G	G	G	G	G	G
5	C		C	C	B					B	F	F	F	F	F	F	F	F
6	C			C	B	B	B	B	B	B	F	F	F	F	F	F		F
7	C	C	C	C	B	D	D	D	D	E	E	E	E	E	E	F		F
8	D	D	D	D	D	D			D	E					E	F		F
9	D							D	D	E	E	E	E	E	E	F		F
10	D	D	D	D	D	D	D	D	H	H	H	H	H	E	E	F	F	F

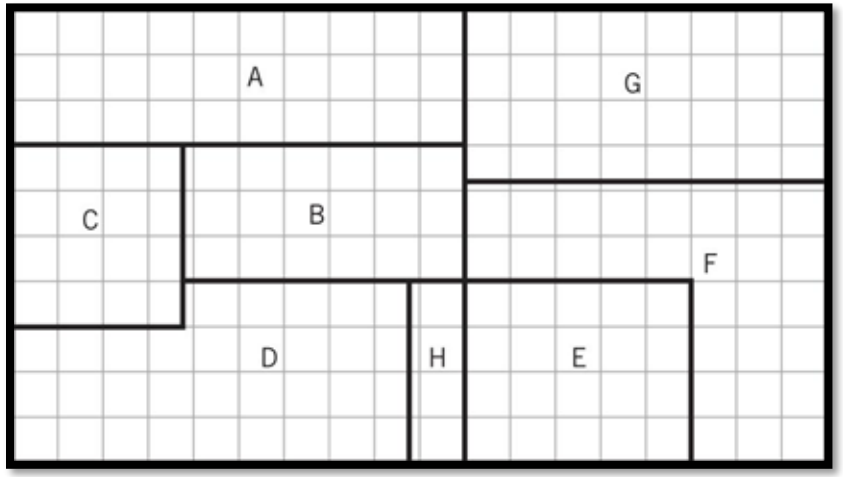
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CRAFT Example

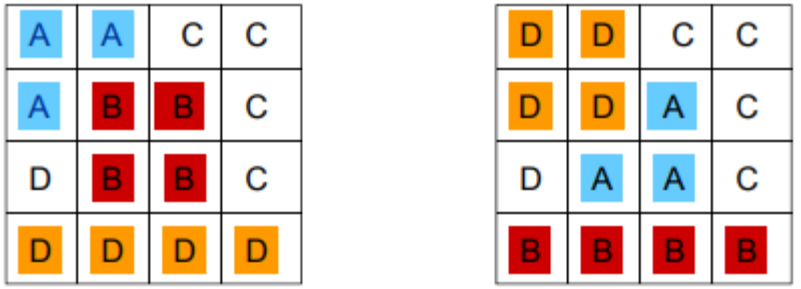
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	A	A	A	A	A	A	A	A	A	A	G	G	G	G	G	G	G	G
2	A									A	G							G
3	A	A	A	A	A	A	A	A	A	A	G	G	G					G
4	C	C	C	B	B	B	B	B	B	B	F	F	G	G	G	G	G	G
5	C		C	C	B					B	F	F	F	F	F	F	F	F
6	C			C	B	B	B	B	B	B	F	F	F	F	F	F		F
7	C	C	C	C	B	D	D	D	D	E	E	E	E	E	E	F		F
8	D	D	D	D	D	D			D	E					E	F		F
9	D							D	D	E	E	E	E	E	E	F		F
10	D	D	D	D	D	D	D	D	H	H	H	H	H	E	E	F	F	F

We can make adjustments so layout is more realistic.
Called "massaging"

Notice that final layout does not fit the unit grid of 20X20 exactly.



Examples for 3-way Exchange



3-way interchange of A, B and D

A → B → D → A

https://www2.isye.gatech.edu/~pinar/teaching/isye3104-fall2012/layout.pdf

CRAFT

Remarks:

- CRAFT estimates the new objective function
  - Does not swap departments first and then calculate
  - Results in estimation error if departments are not equal in area
- Adjacency is a necessary but not sufficient criteria for swapping departments
- Quality of final solution depends on the initial layout
- Use dummy departments
  - to fill extra spaces or building irregularities
  - to evaluate aisle location or obstacles/unusable areas (stairways, elevators, etc.)
- CRAFT is a “steepest descent” algorithm
- It doesn’t look ahead or back
- Therefore, the solution may be locally optimal, not globally optimal
- Because departments can be irregular, the shapes can deteriorate as the number of iterations increase

CRAFT

- Limitations in Exchange: Adjacency is a necessary but not sufficient criteria for swapping departments

- Adjacent departments 2 & 4 gives the lowest objective function value.
- We cannot exchange without splitting department 2.
- Most cases, two adjacent unequal-area departments can be exchanged without splitting.

6	6	6	5	5
6	6	6	5	5
6	6	6	5	4
6	6	6	4	4
2	2	2	2	2
1	1	2	3	3
1	1	2	3	3

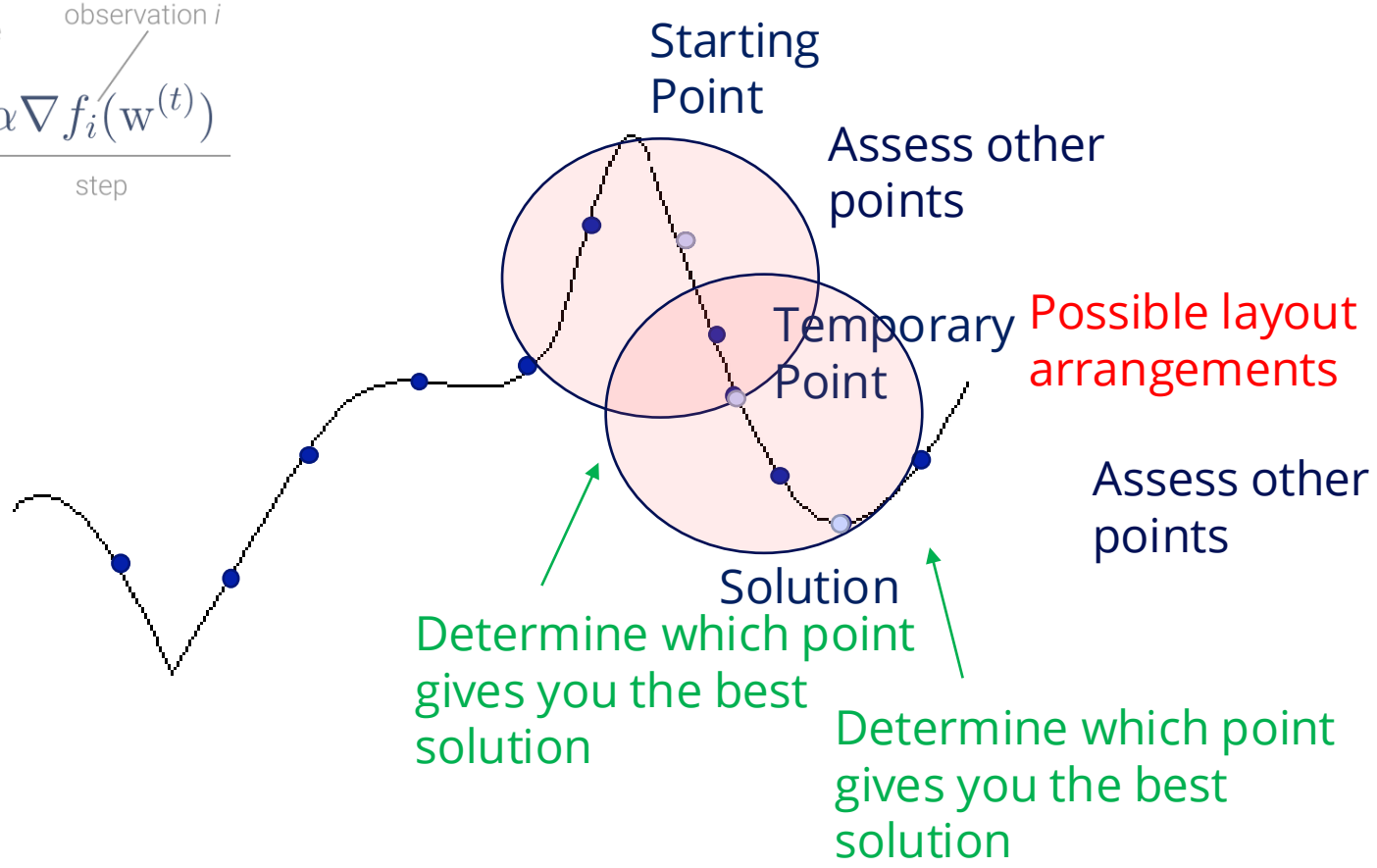
Pair-wise exchange/CRAFT from steepest descent perspective

$$\underbrace{\mathrm{w}^{(t+1)}}_{\text{position of next iteration}} = \mathrm{w} - \underbrace{\alpha \nabla f_i(\mathrm{w}^{(t)})}_{\text{step}}$$

learning rate

observation *i*

position of previous step



Steps in CRAFT

1. Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).

2. Find the cost of the initial layout by multiplying the from-to chart, unit cost matrix and the distance matrix.

3. Improve the layout by performing all-possible two or three-way exchanges (we just consider two-way exchanges here if not stated otherwise).

   - At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.

4. If possible, make adjustments (“Massaging”) for departments layout to be practical.

MICRO-CRAFT (MCRAFT)

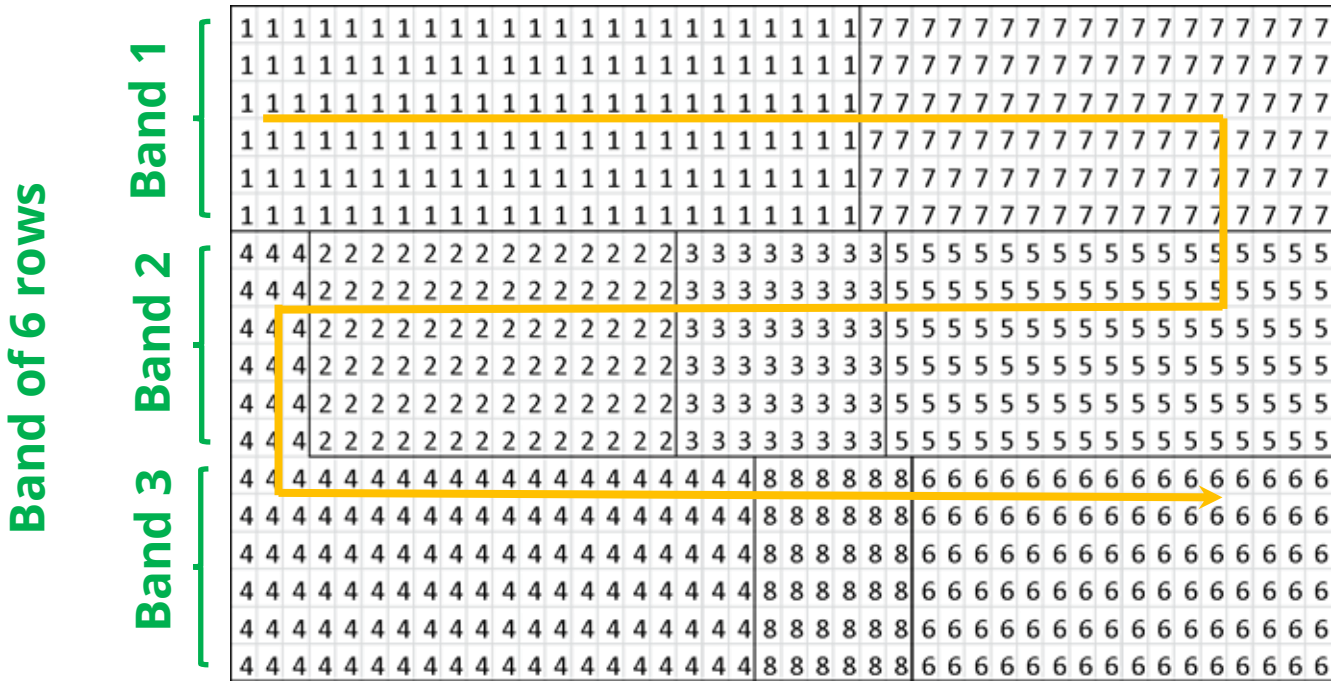
- Can exchange any two departments whether they are adjacent or not (recall that CRAFT require adjacency).
- Shifts other departments “automatically” if the exchanged departments are not adjacent and equal sized.
- Uses “bands” and assigns grid within band to one or more departments
  - Asks for length and width of building, # of bands
  - Computes grid size, number of rows and columns

MCRAFT Example

			FLOW							
Department Name	Area (ft^2)	# of Grids	A	B	C	D	E	F	G	H
1. A:Receiving	12,000	120	0	45	15	25	10	5	0	0
2 B:Milling	8,000	80	0	0	0	30	25	15	0	0
3 C:Press	6,000	60	0	0	0	0	5	10	0	0
4 D:Screw m/c.	12,000	120	0	20	0	0	35	0	0	0
5 E:Assembly	8,000	80	0	0	0	0	0	65	35	0
6 F:Plating	12,000	120	0	5	0	0	25	0	65	0
7 G:Shipping	12,000	120	0	0	0	0	0	0	0	0
8 H:Dummy	2,000	20	0	0	0	0	0	0	0	0

- Set the length equal to 360 and width equal to 200 unit grid; number of bands equal to three.

MCRAFT Example



Forms layout by starting at upper left corner and "sweeping" the bands in a serpentine fashion. ("layout vector" or "fill sequence")

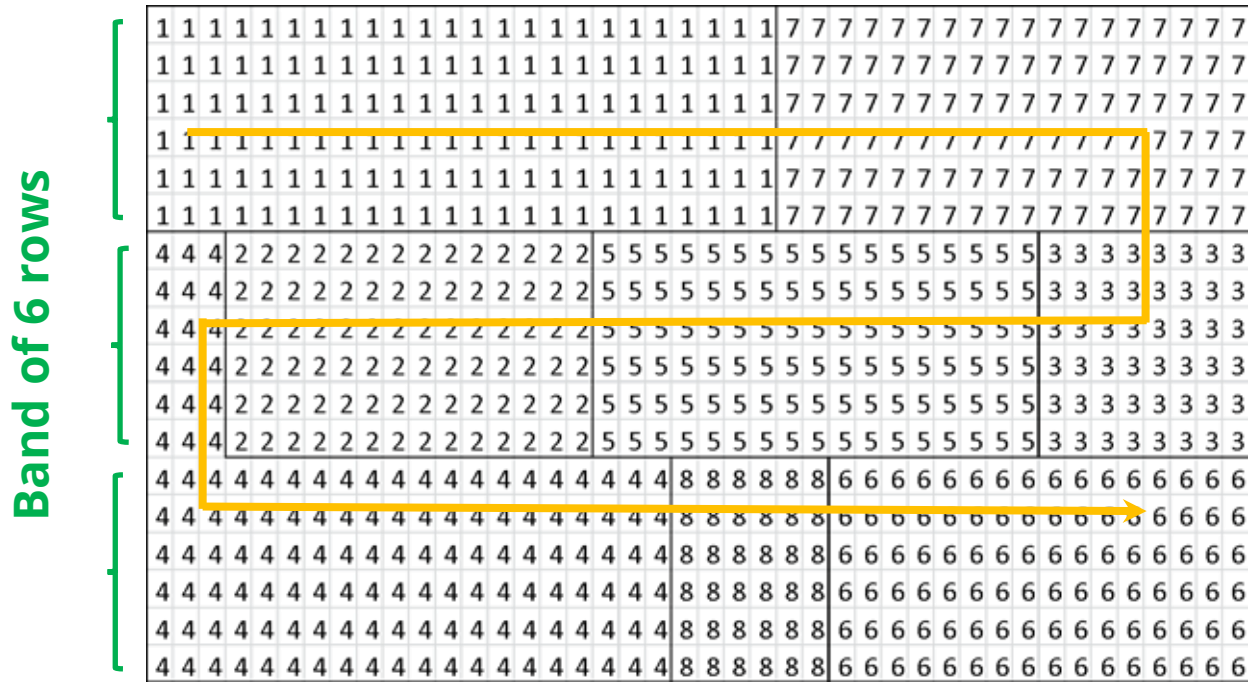
1-7-5-3-2-4-8-6

Iteration1: Identify the best swap between 2 or 3 departments
C-E (3 and 5)

Cost = 59,611.11 units

Non-adjacent departments can be switched.
Non-equal sizes of departments can be switched.

MCRAFT Example

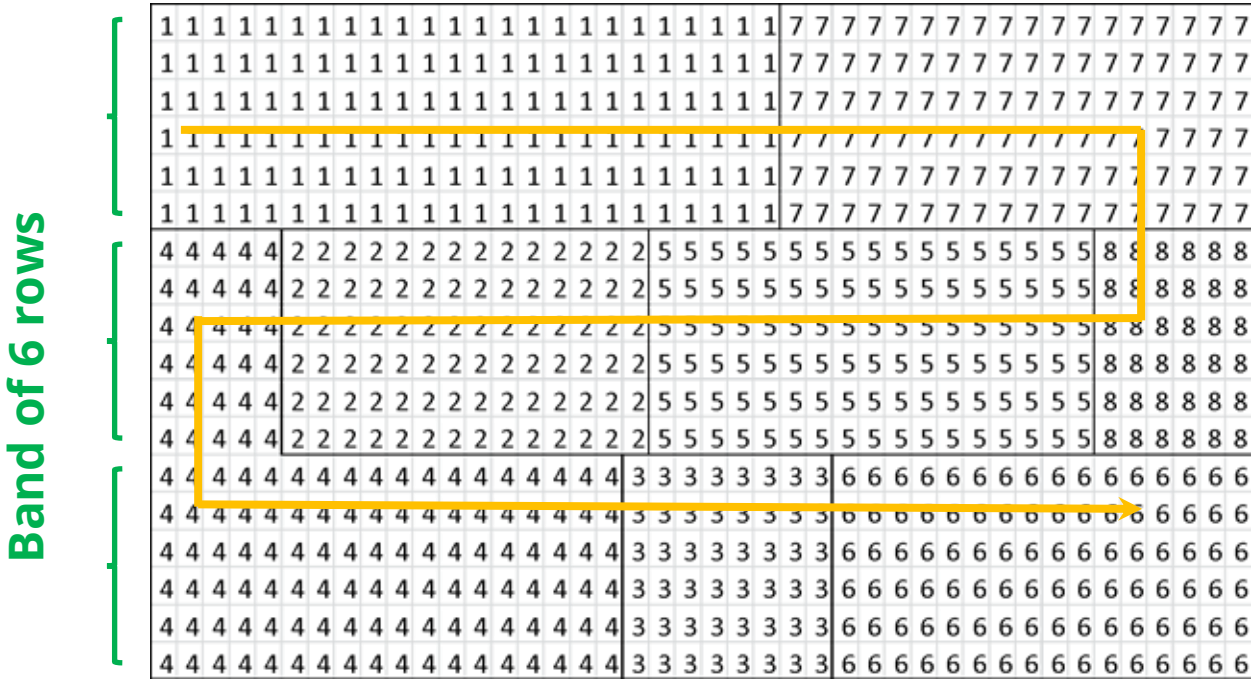


1-7-3-5-2-4-8-6

Iteration2: : Identify the best swap between 2 or 3 departments
C-H (3 and 8)

Cost = 58,083.34 units

MCRAFT Example

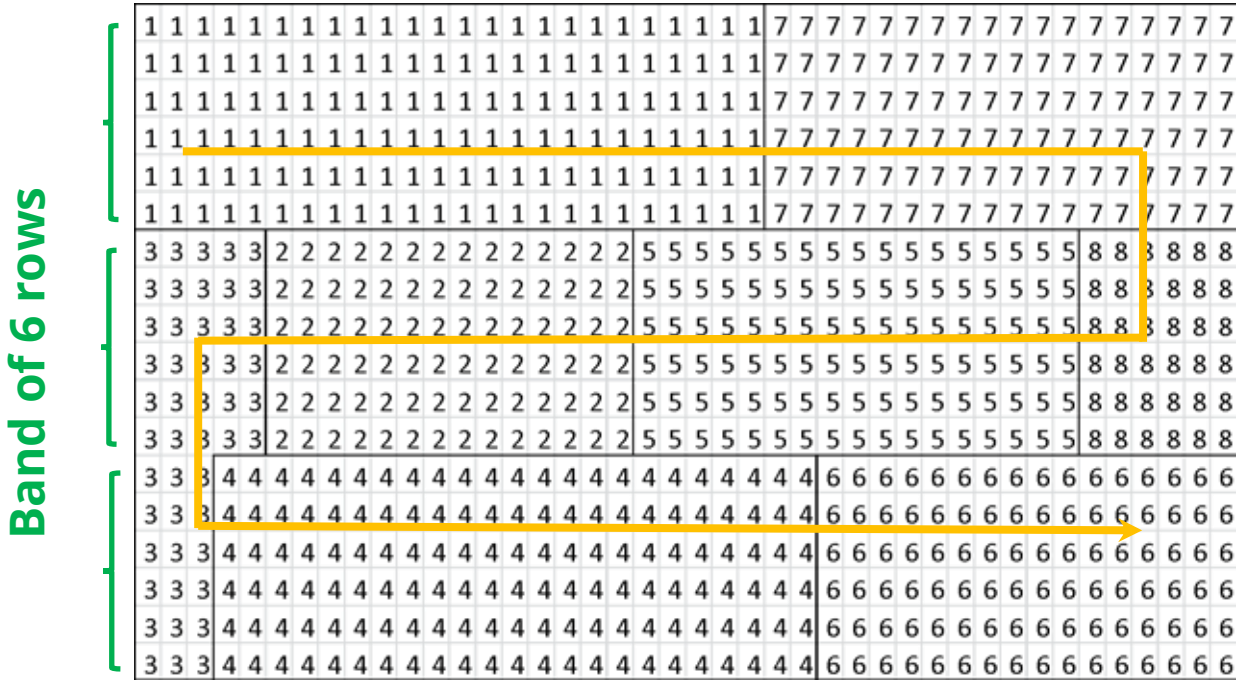


1-7-8-5-2-4-3-6

Iteration3: : Identify the best swap between 2 or 3 departments
C-D (3 and 4)

Cost = 57,483.34 units

MCRAFT Example

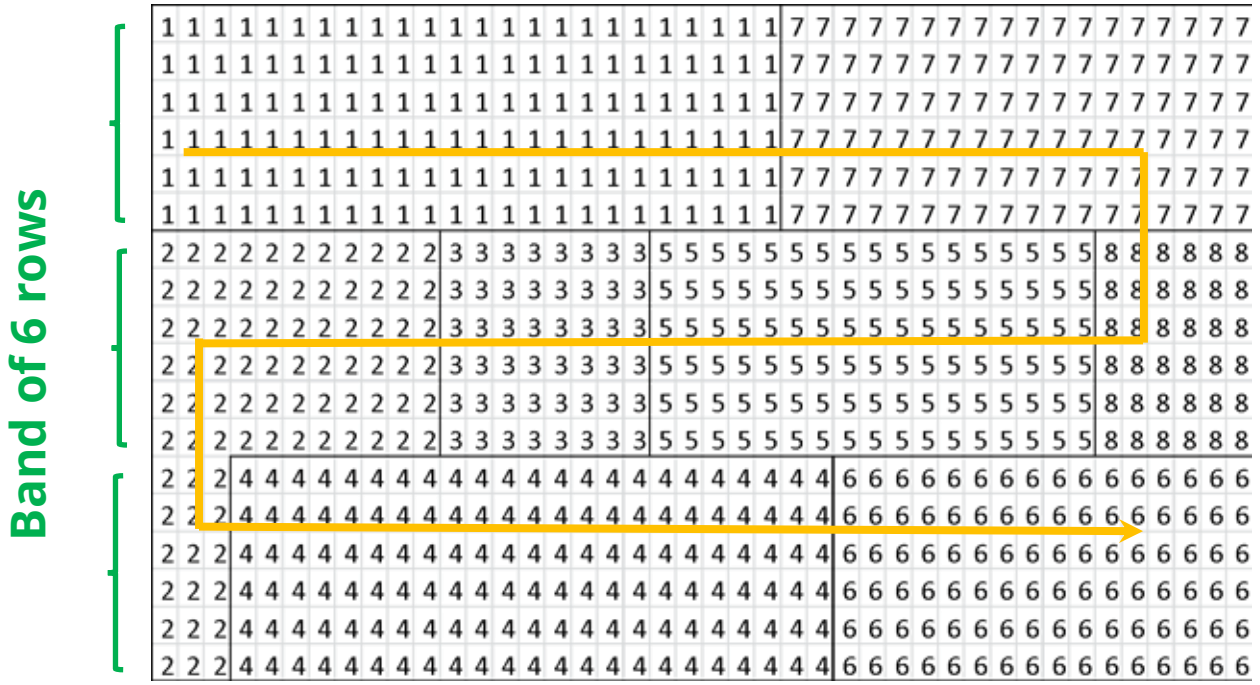


1-7-8-5-2-3-4-6

Iteration4: : Identify the best swap between 2 or 3 departments
B-C (2 and 3)

Cost = 57,333.34 units

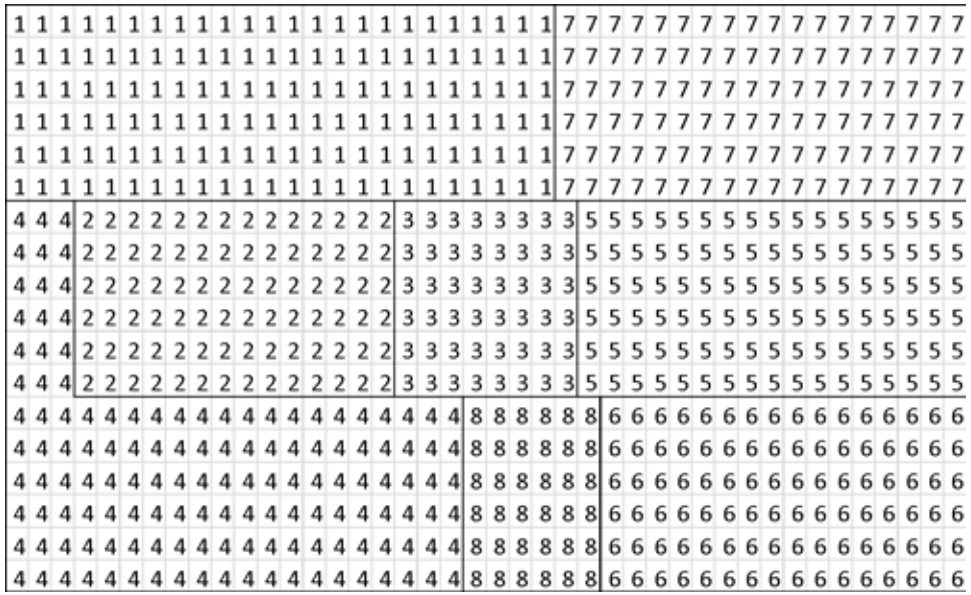
MCRAFT Example



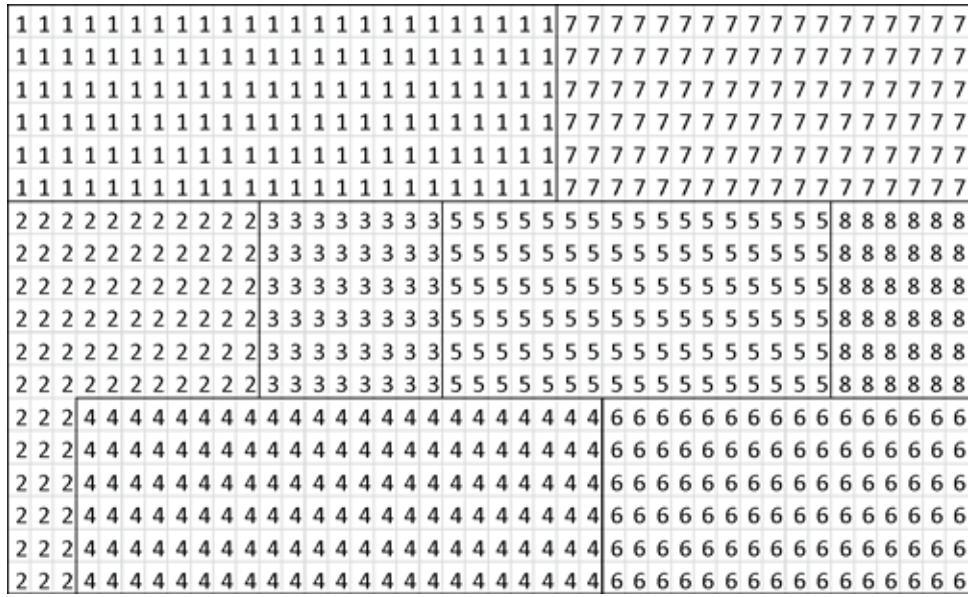
1-7-8-5-3-2-4-6

Iteration 5: Identify the best swap between 2 or 3 departments
No improvement can be made

MCRAFT Example



Cost = 59,611.11 units



Cost = 57,333.34 units

MCRAFT

- Band width is assumed to be the same for all the bands.
- In general, MCRAFT is not as effective as CRAFT in treating obstacles and fixed departments.
  - E.g., the obstacle cannot exceed the band width.

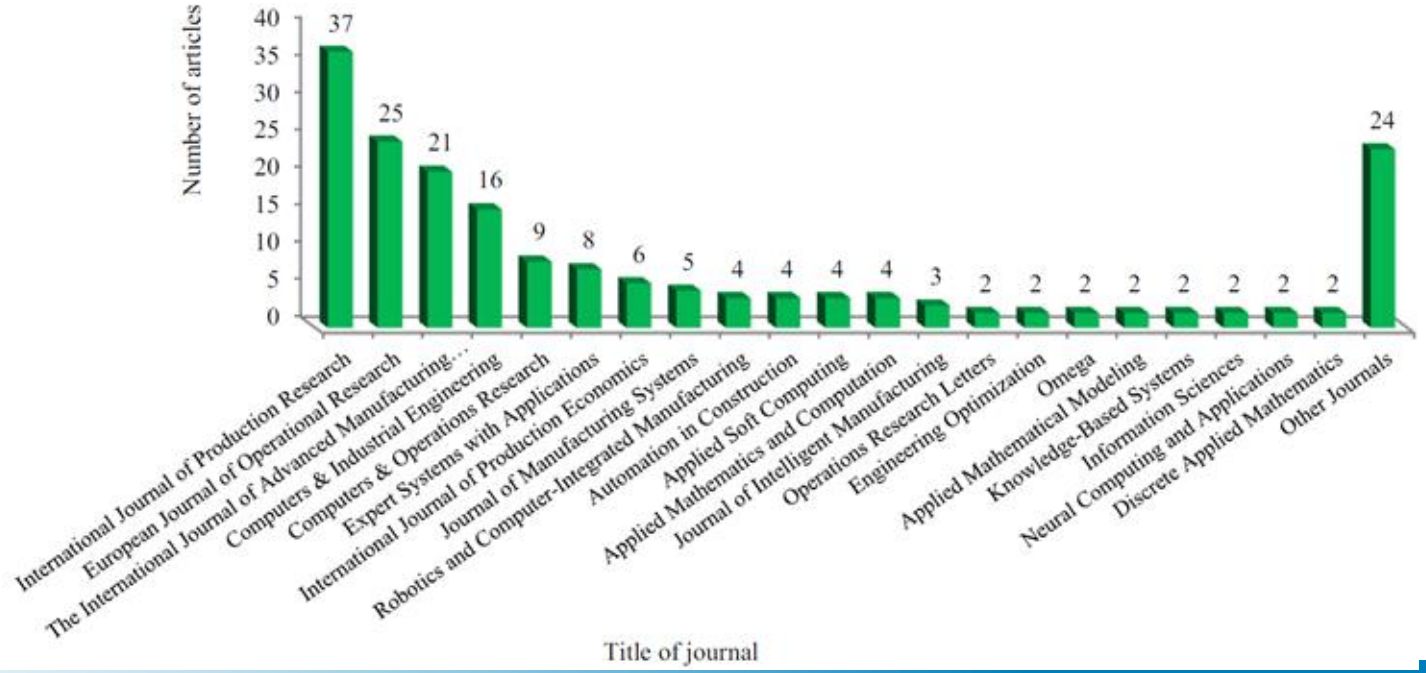
MCRAFT

Algorithm	Space requirement	Representation	Techniques
Pairwise exchange method	Cannot handle		
Graph-based method	Cannot handle		
CRAFT	Can handle	Discrete	grids
MCRAFT	Can handle	Continuous	bands

Algorithm	Objective Function	Type	Restrictions of exchange
Pairwise exchange method	Distance-based	Improvement	Adjacent departments
Graph-based method	Adjacency-based	Construction	
CRAFT	Distance-based	Improvement	Adjacent departments
MCRAFT	Distance-based	Improvement	No restriction

Other Methods?

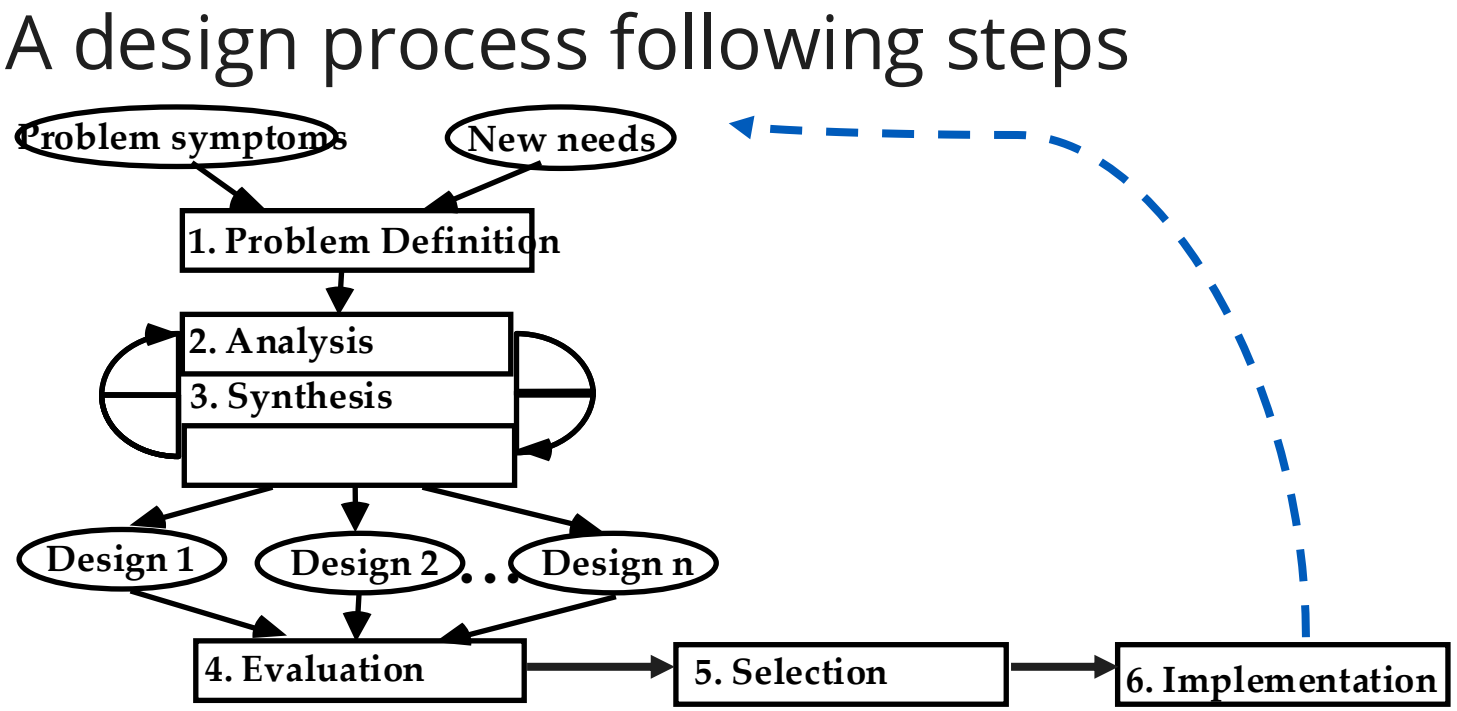
- Read the literature and get the most up-to-date results!
- For instance, by searching “facility layout algorithm review survey” at google scholar
  - A review of intelligent approaches for designing dynamic and robust layouts in flexible manufacturing systems (2012)
  - Classification of facility layout problems: a review study (2018)



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Facility Planning

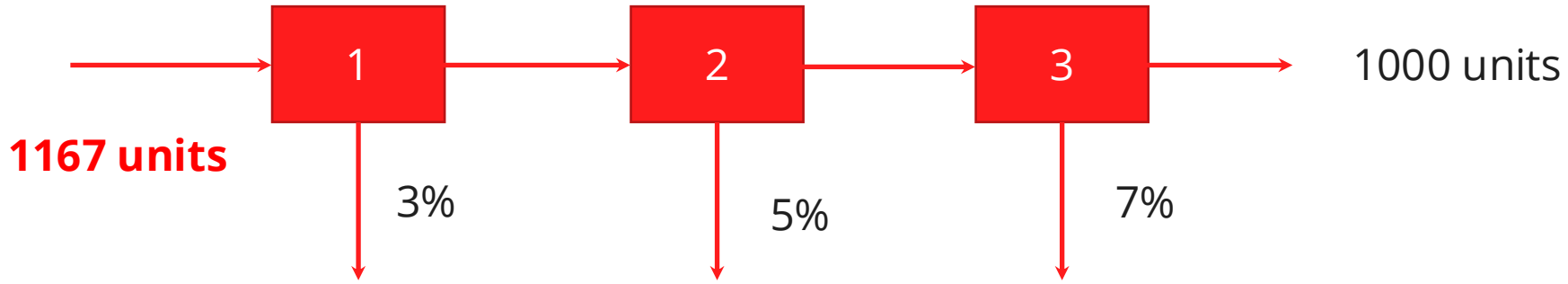
- A design process following steps



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Exercise 1

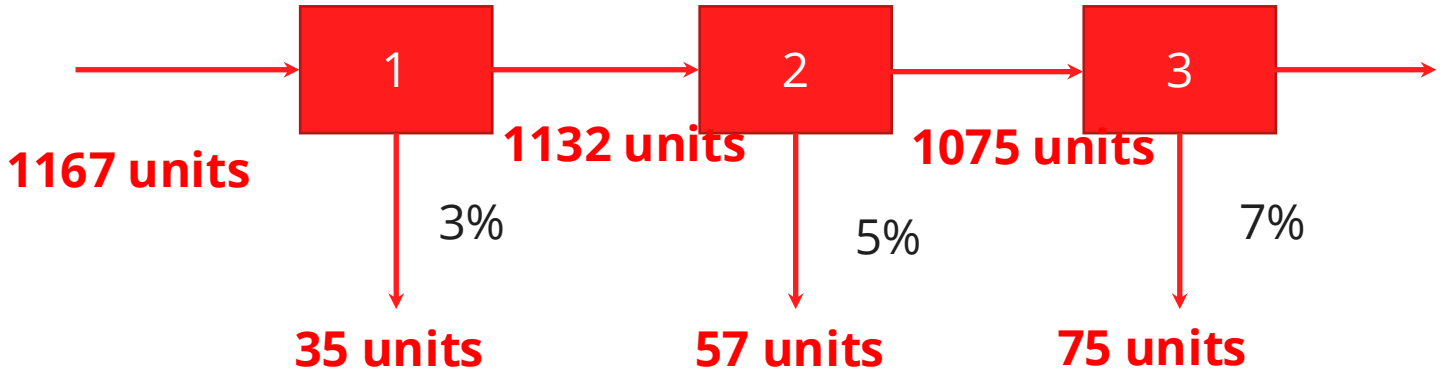
- Exercise 1:
- Consider a simple three-step manufacturing process as illustrated. Assume demand of 1000 units. What is the required input to meet demand?
- Manufacturing Process 1:



Exercise 1

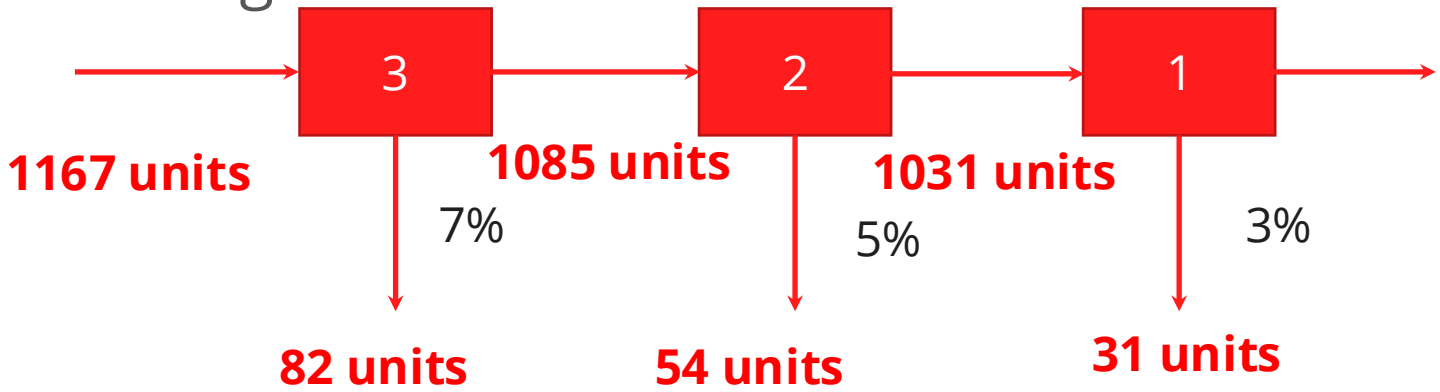
What is the total cost of defects for each manufacturing process? Would there be any difference between the two processes?

- Manufacturing Process 1:



$15*35+$10*57+$5*75 = $1470

- Manufacturing Process 2:



$5*82+$10*54+$15*31 = $1415

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Exercise 2

A company produces a part that requires operation on two types of machines (first A then B). The production planning department forecasts an average of 20 boxes per day. A box of parts has 100 units which are produced in a batch (i.e., lot-size is 100), and a new set-up is required to adjust the machine after each box of parts is produced. Determine the space requirements of the facility with the following data. Assume one shift of 8 hours per day.

Machine	A	B
Processing Time	2 min	1 min
Set-up time/batch	10 min	15 min
Scrap rate	15%	10%
Space per machine	100 ft^2	50 ft^2
Aisle Allowance (% of mc. area)	10%	25%

Step 0. Identify demand

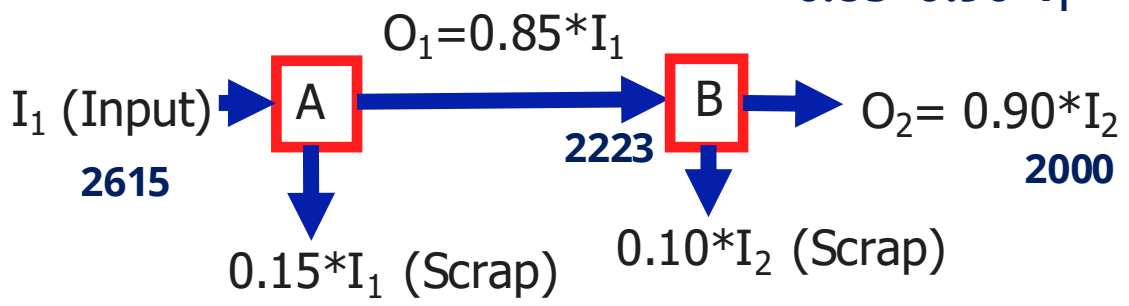
20 * 100 = 2000

Step 1. Identify input amount.

1-1: Identify the process

1.2: Calculate input and scrap

$0.85*0.90*i_1 = 2000$



Exercise 2

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A: 2615 B:2223

Step 2. Determine how many machines are needed

For 1 machine, 2 batches take 7 hours. For the remaining 1 hour, you can set up a batch for 10 minutes, and process 25 units for 50 minute.
225 units per machine per day.

MF (**in terms of units**) = 2615 / 225 = 11.62, 12 machines

For A, one batch takes 10min + 200 = 210 min = 3.5 hours

And we need 26 batches + the last batch with 15

MF (**in terms of time**) = 26.15 * 3.5 / 8=11.44, 12 machines

Exercise 2

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Step 3. Calculate space requirement

A: 12 B: 6

	A	B
# needed	12	6
Total space required	12*100 = 1200	6*50 = 300
Aisle Allowance	1200 * 0.10 = 120	300 * 0.25 = 75
Total Space Requirement	1320	375

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Exercise 3

The ABC Cooling and Heating Company manufactures several different types of air conditioners. Five departments are involved in the processing required for the products. A summary of the processing sequences required for the five major products and the weekly production volumes for the products are shown in the tables below along with the department area. Based on the graph-based construction method, develop a block layout. [Hint: Develop the adjacency graph first]

Product	Process Sequence	Weekly Production
1	ABC	150
2	ABED	200
3	ACE	50
4	ACBE	200
5	ADE	250

Department	Area (ft^2)
A	1500
B	1500
C	1000
D	2000
E	2000

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